

Day 16: Motion Mask Cleanup & Stabilization

Outcomes:

- Understand why raw motion masks are unreliable
- Learn how noise, holes, and shadows appear in motion detection
- Know how masks are cleaned before contour detection
- Be able to reason about stable vs accurate detection

Why raw motion masks fail

Background subtraction outputs a foreground probability, not a clean object.

This causes:

- scattered white pixels (noise)
- broken object shapes
- shadow regions marked as foreground
- flickering detections

This is expected behavior.

Noise removal

Small moving pixel groups are meaningless.

They usually come from:

- camera sensor noise
- lighting flicker
- background model updates

Conceptually, noise is removed by ignoring small regions and cleaning isolated pixels.

Filling gaps inside objects

Foreground objects often appear hollow.

This happens because:

- motion is not uniform across the object
- some regions momentarily match the background

Filling gaps creates solid, usable shapes for contour detection.

Shadow handling

Shadows move, but they are not objects.

Many background subtractors label shadows differently.

Conceptually:

- shadows are filtered out by intensity-based rejection
- only strong foreground pixels are kept

Shape stabilization

Even after cleanup, object edges fluctuate.

Stability is achieved by:

- smoothing shapes over time
- accepting slightly larger regions for consistency

In vision systems:

A stable detection is more valuable than a perfect one.

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